INDIAN SCHOOL OF BUSINESS · INSTITUTE OF DATA SCIENCE

CAPSTONE PROJECT: DATA NOTE

Project: Demand Prediction by SKU: Forecasting Lumpy Room Air-Conditioner Demand for Depot Replenishment
Sponsor: Deloitte Touche Tohmatsu India LLP | Cohort: AMPBA Class of 2026S | Date: October 2026

Sponsor & Client Context	Deloitte Touche Tohmatsu India LLP (advising a leading Indian room air-conditioner manufacturer)
Source Data Extraction	SAP ERP billing extract spanning 16 July 2022 to 30 June 2024 (103 weekly billing cycles; 82,034 rows)
Data Sensitivity Level	CONFIDENTIAL / RESTRICTED (Commercial client billing data shared under sponsor NDA for ISB evaluation)
Target Demand Grain	Weekly billing quantity for 70 outdoor-unit (ODU) models across 5 regional sales depots (BLR, COK, MAA, SBD, SBD1)
Sponsor Approval Status	Data-handling protocol and academic usage approved by Deloitte in August 2026

1. Data Description & Provenance

The dataset comprises transactional ERP billing lines from the manufacturer's South India operations. Each record captures billing date, material code, sales office code, billing quantity, equipment segment (inverter vs. fixed-speed), star rating, and tonnage. Following comprehensive forensic audits, indoor-unit lines (IDUs) were set aside (demand is strictly 1:1 with outdoor units) and exact duplicate rows in Q1 FY25 were removed, establishing an empirical data mart of 66,336 positive ODU lines and 698,904 shipped units.

2. Anonymization & Confidentiality Safeguards

In strict compliance with ISB Honor Code and sponsor confidentiality protocols, the following safeguards were applied:

- Client Manufacturer Masking: The corporate entity and brand name of the AC manufacturer are fully anonymized.
- Personally Identifiable Information (PII) Exclusion: The extract contained names of individual retail buyers and dealer names. All customer/dealer names were strictly quarantined, omitted from feature engineering and models, and appear in zero tables or charts.
- Commercial Pricing Suppression: No invoice values, unit prices, billing margins, or revenue figures are disclosed; all analyses and replenishment policies operate on unit volumes.
- Aggregation Thresholds: All published exhibits and summary tables are aggregated to SKU × branch × week level.

3. Code Repository & Storage Security

The analytical codebase and Streamlit cockpit are maintained in a secure private GitHub repository (https://github.com/saiprash711/ISB_Capstone_Project). All raw billing files, client extracts, and intermediate pickles are excluded from version control via .gitignore. Reproduction instructions utilize synthetic or sanitized mock files.

Submitted on behalf of the Capstone Project Team:
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